Emeka Odiwe

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Education

B.Sc. Biomedical Engineering Sep 2023 – Present
University of Lagos (UNILAG)
- Current CGPA: 4.59

Certificates

- Certificates

Projects & Research

CodeAlpha Internship - Credit Scoring ML System Oct 2025 – Nov 2025
Role: ML Engineer
- Built an end-to-end ML pipeline evaluating **4 supervised learning models**: Logistic Regression, Random Forest, Gradient Boosting, and SVC.
- Performed data preprocessing, exploratory data analysis (EDA), feature encoding, and class-balancing techniques to improve model learning.
- Evaluated models using accuracy, ROC-AUC, precision/recall, and confusion matrices to compare performance and identify the most reliable predictor.
- Applied ML decision-making principles to financial risk assessment and model optimization.

Brain Tumor Classification & Segmentation (BraTS) (in progress) Aug 2025 – Present
Role: Computer Vision Engineer
- Developing pipelines for brain tumor detection and segmentation using CNN models and U-Net–based architectures.
- Implemented preprocessing workflows including multi-channel label conversion, intensity normalization, augmentation, and sliding-window inference using MONAI and Torchio.
- Evaluating model performance and generalization using SwinUNETR to predict Tumors based on the BraTS dataset.
- Working with custom datasets and Python-based tooling to refine training efficiency, reproducibility, and medical imaging model deployment.

NATALINK Jul 2025 – Jul 2025
Role: Computer Vision Engineer : Health Sector
- Designed and implemented a deep-learning model to predict fetal head–pubic symphysis positioning using ultrasound images.
- Utilized YOLOv8 for landmark detection and OpenCV for image preprocessing, measurement extraction, and visualization.
- Improved diagnostic accuracy by automating key anatomical assessments relevant to labor monitoring (head of baby, pubic symphysis of mother).

SPARK Academy — AI in MRI (Research) Feb 2025 – Aug 2025
Role: Python Programmer / Research Trainee
- Gained foundational knowledge of brain tumor types and their clinical relevance.
- Studied how modern AI/ML models (including CNN-based architectures) are trained to detect and segment tumors on MRI scans.
- Learned the different MRI modalities used in brain imaging, including T1, T2, T1-CE, FLAIR, and Segmentation (SEG) maps, and how each contributes to tumor analysis.
- Built hands-on skills in Python for data handling, preprocessing, and understanding AI pipelines for medical imaging.

BeyondAI / Research Stage MLP vs KAN Comparison Sep 2024 – Jan 2025
Role: Beginner ML Engineer
- Conducted a structured literature review covering neural network architectures, activation functions, and training behaviours.
- Designed and ran comparative experiments using 2 classification datasets: Wine Quality and Breast Cancer.
- Implented and evaluated MLP and KAN models performance using accuracy, loss curves, and generalization metrics to analyze strengths and limitations of each approach.
- Documented findings and implemented all experiments in a reproducible codebase.
- Notion Link

Med-ID Jun 2026 – Present
Role: AI Engineer
Med-ID is an AI-powered Portable Health Record (PHR) prototype that allows secure, doctor's access to medical records across healthcare facilities. It was built majorly for patients requiring emergencies at the other side of the country without their details. The system allows for emergency "break-glass" access for verified healthcare providers, AI-generated clinical summaries to hasten decision-making during emergencies, and role-based access control to protect patient privacy. A functional Streamlit MVP was built using OPENCODE and demonstrated during a national innovation hackathon, where we emerged as winners. The project is currently being expanded toward a more production-ready solution.

Leadership & Volunteer

Training Team: Machine Learning Trainer Nov 2025 – Present
BMESA Hub (Biomedical Engineering Association) UNILAG, Lagos
- Supported onboarding of new members into AI/ML programs.
- Assisted in delivering **introductory ML training sessions and coding workshops**.
- Mentoring beginners on ML fundamentals, Python workflows, and project structure.

Volunteer Member - Welfare Team Nov 2025 – Present
Data Community UNILAG UNILAG, Lagos
- Providing support to members during community initiatives and events.
- Coordinating team activities to ensure smooth participation and engagement.
- Assisting in maintaining an organized, supportive environment for all participants.

Tutor / Mentor Present
Self employed
- Provides academic guidance to students through one-on-one and group tutoring sessions.
- Teaching students key concepts across various subjects, helping them improve performance.
- Support learners with assignments, study strategies, and personalized learning plans.

Assistant Team Lead Feb 2025 – Aug 2025
SPARK Team AFIRIKAI Lagos,Nigeria
- Coordinated workflow for a **multi-member research team** during MRI tumor segmentation tasks.
- Managed internal deadlines and supported model experimentation during the hackathon phase.
- Assisted teammates with preprocessing pipelines and experiment organization.

Technical Skills

Languages & Tools ●●●○○
Python, PyTorch, NumPy, Pandas, scikit-learn, Git

Medical Imaging Tools ●●○○○
MONAI, custom U-Net, SwinUNETR, Torchio.

Other
Research methods, literature review, poster presentation, tutoring/teaching

Competitions & Awards

Silver Medal, Departmental Chess Competition May 2024

Winner, NIMECHE Innovation Competition Aug 2025

First Runner-Up, Biodrive 2.0 Innovation Competition Jul 2025

Winner, 234 Ai Hackathon Jun 2026

Portfolio Links (some in progress)

github
https://github.com/meeks627

Natalink
Google Colab Natalink

Brain Tumor Segmentation Aug 2025 – Present
BraTS Lagos

Medical ID Jun 2026 – Present
github repo Yaba, Lagos
demo video